MediSafe AI — Clinical Decision Support Report

AI-Powered Disease Risk Classification & Medication Safety Decision Support

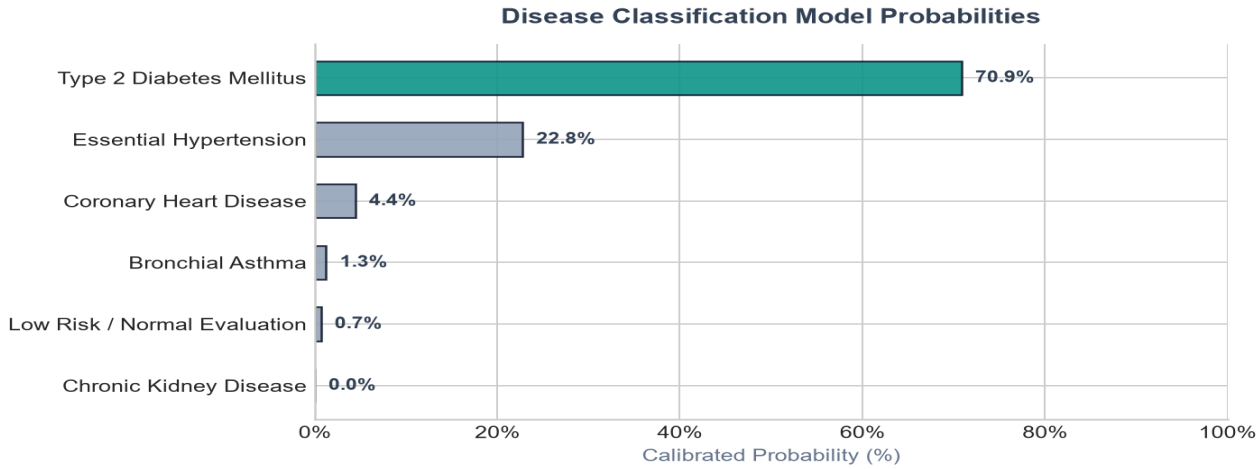
CLINICAL DECISION SUPPORT NOTICE: This document contains algorithmic evaluations for clinical guidance only. It does NOT constitute a confirmed medical diagnosis and cannot prescribe, modify, or discontinue medications. Professional verification by a licensed physician or pharmacist is required.

1. Patient Profile & Clinical Parameters

Patient ID:	P0001	**Full Name:**	Arthur Pendelton
Age / Gender:	58 yrs / Male	**BMI:**	29.4 kg/m²
Blood Pressure:	142.0 mmHg	**Heart Rate:**	84.0 bpm
Fasting Glucose:	168.0 mg/dL	**HbA1c:**	7.8%
Cholesterol:	235.0 mg/dL	**Smoking Status:**	Yes
Reported Symptoms:	Frequent urination, excessive thirst, blurred vision, mild fatigue	**Known Allergies:**	Penicillin, Sulfa drugs

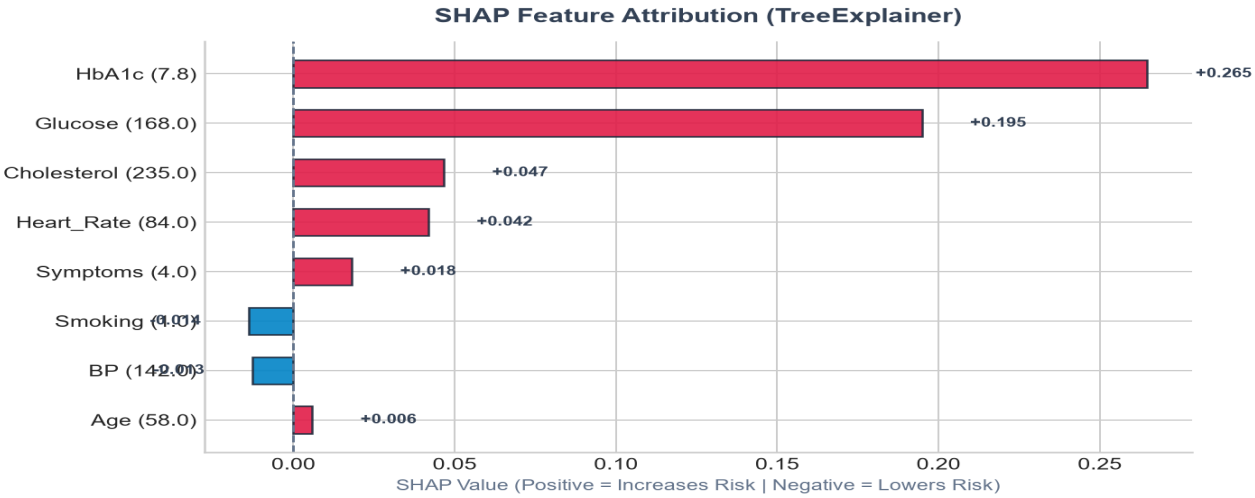
2. Random Forest Disease Classification

Primary Prediction: **Type 2 Diabetes Mellitus** (Model Calibrated Probability: **70.9%**) | **Assigned Safety Level:** **HIGH**



3. Explainable AI (SHAP TreeExplainer Analysis)

The model's classification toward 'Type 2 Diabetes Mellitus' was most heavily influenced by: HbA1c (Increases Risk), Glucose (Increases Risk), Cholesterol (Increases Risk). Features with positive SHAP values shifted probability toward this classification.



4. Medication Safety, OCR & Drug-Drug Interactions

Active Regimen: Metformin 850mg, Lisinopril 10mg
Prescription OCR Detected: None detected
Allergy Status: No active allergy contraindications flagged.

Drug Pair	Severity	Clinical Mechanism / Interaction	Evidence
Metformin 850mg + Lisinopril 10mg	**Moderate**	ACE inhibitors like Lisinopril may increase hypoglycemic sensitivity and enhance blood glucose lowering effect of Metformin. Regular glucose and renal monitoring advised.	Clinical Pharmacology Database (DrugBank/Lexicomp)

5. OpenFDA Pharmacovigilance Surveillance Evidence

Medicine	Total FAERS Reports	Prevalent Adverse Reactions
Metformin	798,429	NAUSEA (30,192), DIARRHOEA (28,014), BLOOD GLUCOSE INCREASED (27,174)
Lisinopril	625,156	FATIGUE (21,079), NAUSEA (19,678), DRUG INEFFECTIVE (18,973)

FDA adverse-event reports are reports of events and do not by themselves establish causality.

6. Multi-Agent Synthesis & Clinical Recommendations

• **Disease Analysis Agent:** Evaluated Random Forest classification: The model assigned highest probability to 'Type 2 Diabetes Mellitus' with a calibrated model probability of 70.9%. Major drivers identified by explainability analysis: HbA1c (SHAP: +0.2647), Glucose (SHAP: +0.1948), Cholesterol (SHAP: +0.0466).

• **Medication Safety Agent:** Medication reconciliation completed across 2 current medication(s) and 0 prescription-extracted entity/entities. 0 allergy alert(s) and 1 drug-drug interaction warning(s) flagged.

• **FDA Evidence Agent:** Queried OpenFDA Adverse Event Reporting System (FAERS). Analyzed post-marketing surveillance for 2 active ingredient(s) totaling 1,423,585 historical event reports. Most prevalent reported reactions: Metformin: NAUSEA, DIARRHOEA; Lisinopril: FATIGUE, NAUSEA.

• **Risk Assessment Agent:** Multi-source clinical synthesis categorizes current patient risk profile as: HIGH.

• **Report Agent:** Multi-Agent Clinical Decision Support Summary: Overall safety level evaluated as 'HIGH'. Disease Analysis Agent reports high association with primary predicted classification. Medication Safety Agent highlighted 1 actionable pharmacological alert(s). FDA Evidence Agent verified post-marketing surveillance profiles. Professional medical review is required prior to enacting clinical interventions.

Mandatory Professional Verification:

This report is generated for qualified clinical personnel. No dosage adjustments, additions, or cancellations should occur based purely on algorithmic outputs. Sign below after comprehensive clinician review.

Reviewed by Physician / Pharmacist:
_____

Signature: _____

Date: 2026-09-10 16:05

License ID: _____